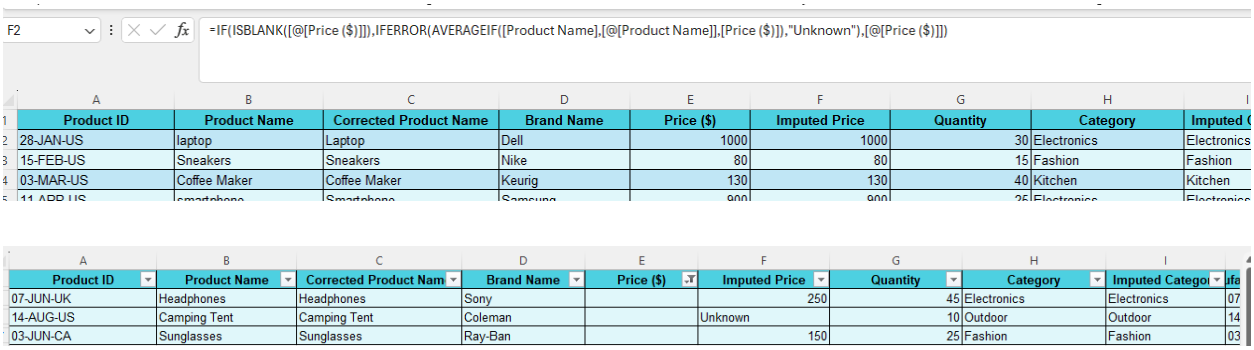
Excel Assignment 2 – Data Exploration

Tasks: 1.

Handling Missing Values:

*** Check for missing values in the 'Price' column. How would you handle products with missing price information?**



F2 =IF(ISBLANK([@[Price ($)]]),IFERROR(AVERAGEIF([Product Name],[@[Product Name]],[Price ($)]),"Unknown"),[@[Price ($)]])

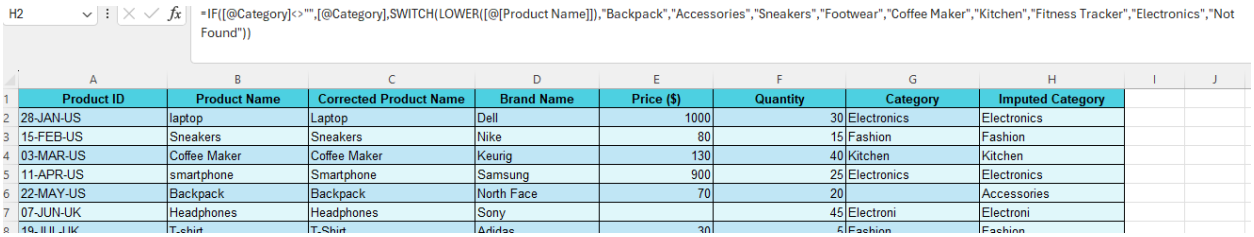
Product ID	Product Name	Corrected Product Name	Brand Name	Price ($)	Imputed Price	Quantity	Category	Imputed (
28-JAN-US	laptop	Laptop	Dell	1000	1000	30	Electronics	Electronics
15-FEB-US	Sneakers	Sneakers	Nike	80	80	15	Fashion	Fashion
03-MAR-US	Coffee Maker	Coffee Maker	Keurig	130	130	40	Kitchen	Kitchen
11-APR-US	smartphone	Smartphone	Samsung	900	900	25	Electronics	Electronics

Product ID	Product Name	Corrected Product Nam	Brand Name	Price ($)	Imputed Price	Quantity	Category	Imputed Categor	afe
07-JUN-UK	Headphones	Headphones	Sony		250	45	Electronics	Electronics	07
14-AUG-US	Camping Tent	Camping Tent	Coleman		Unknown	10	Outdoor	Outdoor	14
03-JUN-CA	Sunglasses	Sunglasses	Ray-Ban		150	25	Fashion	Fashion	03

Summary: Handling Missing Values in the Price Column

I used the IF, ISBLANK, AVERAGEIF, and IFERROR functions to identify missing prices and replace them with the average price of the same product. If no average price is available, the formula displays "Unknown". Existing prices remain unchanged. This helps maintain data accuracy and consistency.

***If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.**



H2 =IF([@Category]<>"",[@Category],SWITCH(LOWER([@[Product Name]]),"Backpack","Accessories","Sneakers","Footwear","Coffee Maker","Kitchen","Fitness Tracker","Electronics","Not Found"))

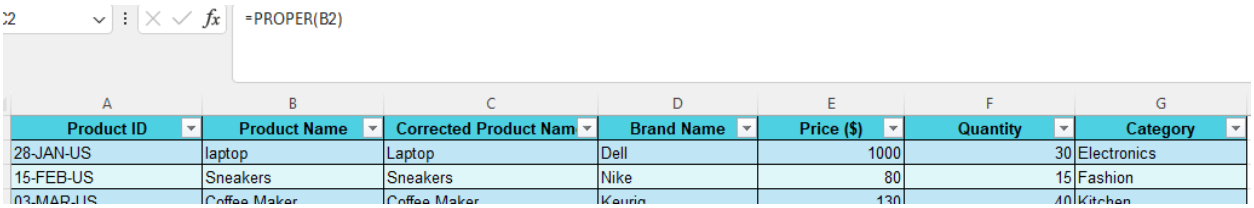
Product ID	Product Name	Corrected Product Name	Brand Name	Price ($)	Quantity	Category	Imputed Category
28-JAN-US	laptop	Laptop	Dell	1000	30	Electronics	Electronics
15-FEB-US	Sneakers	Sneakers	Nike	80	15	Fashion	Fashion
03-MAR-US	Coffee Maker	Coffee Maker	Keurig	130	40	Kitchen	Kitchen
11-APR-US	smartphone	Smartphone	Samsung	900	25	Electronics	Electronics
22-MAY-US	Backpack	Backpack	North Face	70	20		Accessories
07-JUN-UK	Headphones	Headphones	Sony		45	Electroni	Electroni
19-JUL-UK	T-shirt	T-Shirt	Adidas	30	5	Fashion	Fashion

Summary: Handling Missing Values in the Category Column

I used the IF, SWITCH, and LOWER functions to identify missing categories and assign appropriate categories based on the product name. Existing categories remain unchanged. If a product name does not match the predefined list, the formula displays "Not Found". This helps maintain data consistency and reduces missing values.

2. Correcting Inconsistent Data:

● Identify any inconsistent text formats present in the "Product Name" column. ● Identify any typos present in the "Category" column.

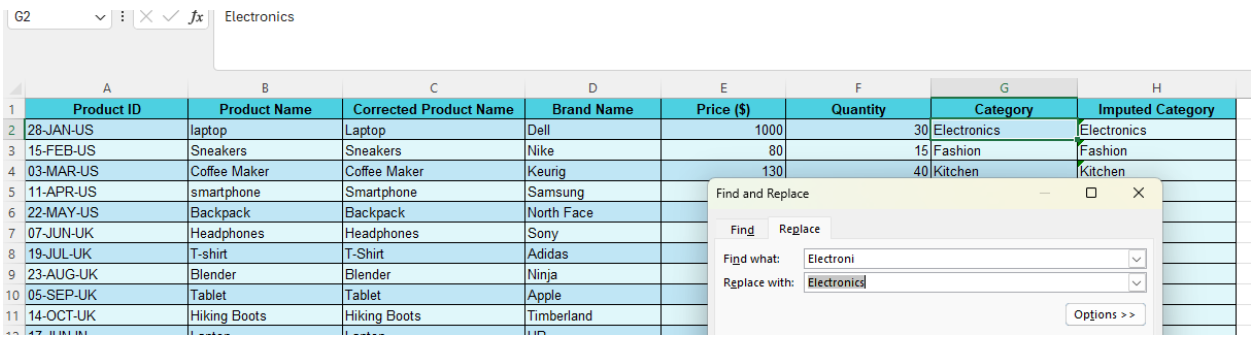


Product ID	Product Name	Corrected Product Nam	Brand Name	Price ($)	Quantity	Category
28-JAN-US	laptop	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Coffee Maker	Keurig	130	40	Kitchen

Summary: Correcting Inconsistent Text Formats and Typos

I used the PROPER function to standardize the text format in the Product Name column by converting the first letter of each word to uppercase. I also checked the Category column for spelling mistakes and corrected any typos to maintain consistency and accuracy in the dataset.

● Use the find and replace function to standardize the text formats in the "Product Name" column and fix any typos or misspellings in the "Category" column.



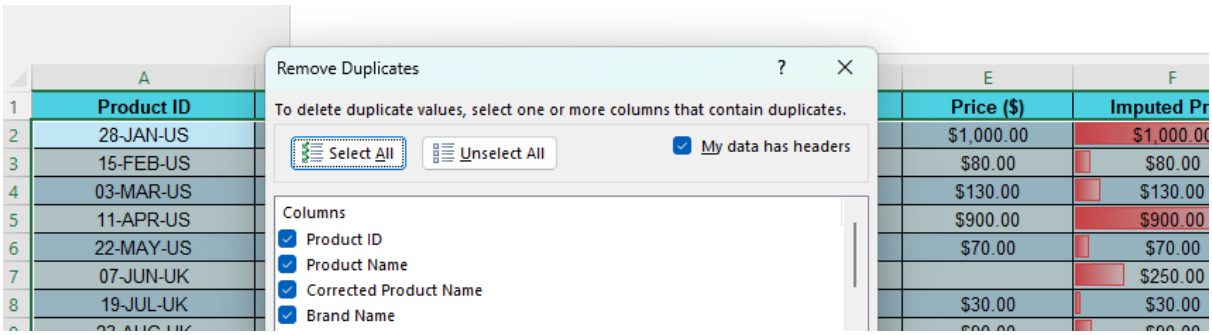
Product ID	Product Name	Corrected Product Name	Brand Name	Price ($)	Quantity	Category	Imputed Category
28-JAN-US	laptop	Laptop	Dell	1000	30	Electronics	Electronics
15-FEB-US	Sneakers	Sneakers	Nike	80	15	Fashion	Fashion
03-MAR-US	Coffee Maker	Coffee Maker	Keurig	130	40	Kitchen	Kitchen
11-APR-US	smartphone	Smartphone	Samsung				
22-MAY-US	Backpack	Backpack	North Face				
07-JUN-UK	Headphones	Headphones	Sony				
19-JUL-UK	T-shirt	T-Shirt	Adidas				
23-AUG-UK	Blender	Blender	Ninja				
05-SEP-UK	Tablet	Tablet	Apple				
14-OCT-UK	Hiking Boots	Hiking Boots	Timberland				

Summary: Standardizing Product Names and Correcting Category Typos

I used the Find and Replace function in Excel to standardize inconsistent text formats in the Product Name column and correct spelling mistakes or typos in the Category column. This helps improve data consistency, accuracy, and quality.

3. Removing Duplicates:

● Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.

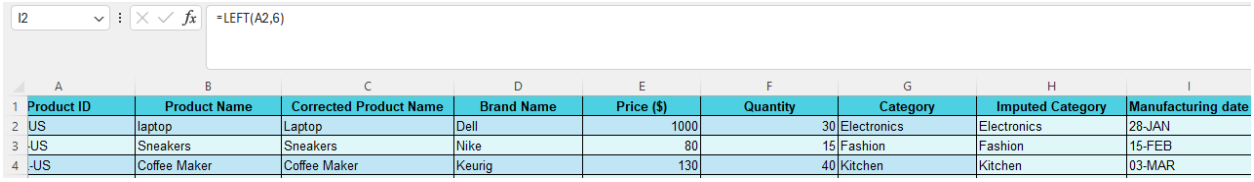


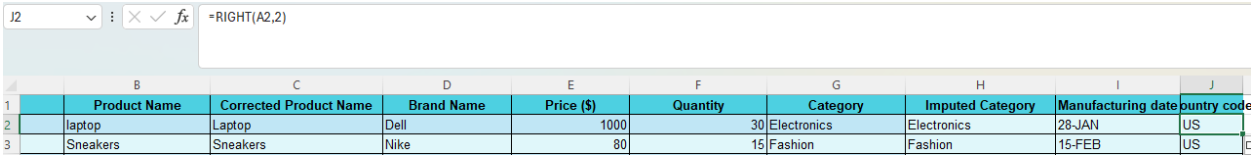
Summary: Identifying and Removing Duplicate Rows

I used the Remove Duplicates feature in Excel to identify and remove duplicate rows based on all columns in the dataset. This ensures each record is unique and helps maintain data accuracy and consistency.

4. Splitting and Merging Data:

● Split the "Product ID" column into two separate columns for " Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.

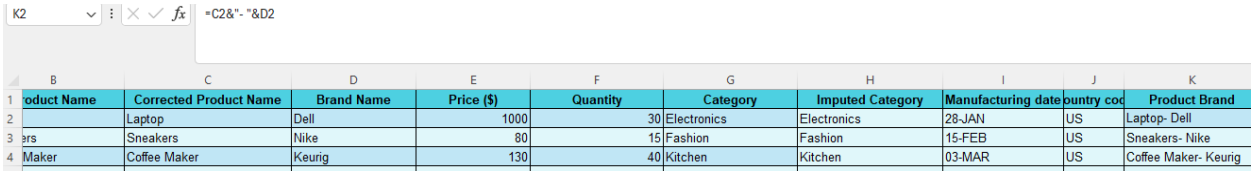




Summary: Splitting Product ID into Separate Columns

I used the LEFT and RIGHT functions in Excel to split the Product ID column into two separate columns: Manufacturing Date and Country Code. I removed unnecessary characters using the appropriate formulas to extract the required information. This helps organize the data and makes it easier to analyze.

● Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".



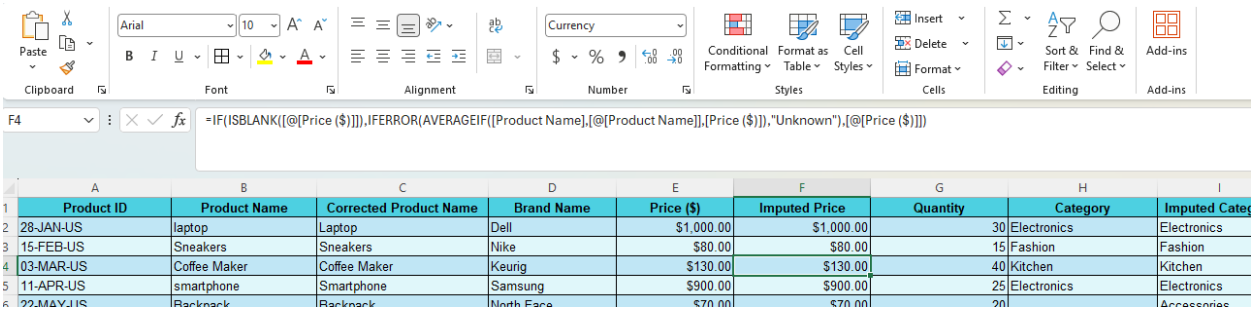
	B	C	D	E	F	G	H	I	J	K
1	oduct Name	Corrected Product Name	Brand Name	Price ($)	Quantity	Category	Imputed Category	Manufacturing date	ountry cod	Product Brand
2		Laptop	Dell	1000	30	Electronics	Electronics	28-JAN	US	Laptop- Dell
3	ers	Sneakers	Nike	80	15	Fashion	Fashion	15-FEB	US	Sneakers- Nike
4	Maker	Coffee Maker	Keurig	130	40	Kitchen	Kitchen	03-MAR	US	Coffee Maker- Keurig

Summary: Merging Brand Name and Product Name

I used the & operator (concatenation) in Excel to combine the **Brand Name** and **Product Name** columns into a single column named **Product Brand**. This helps organize product information into one column and improves data readability.

5. Number Formatting:

● Format the data type of the "Price" column to a currency format.

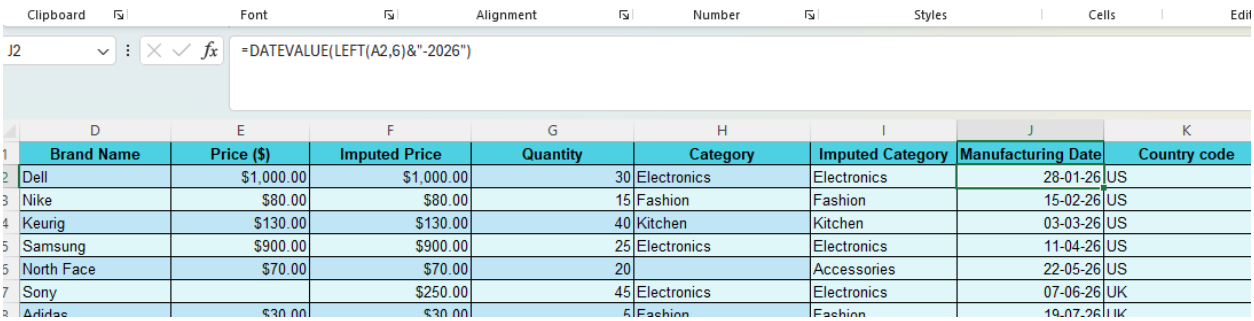


	A	B	C	D	E	F	G	H	I
1	Product ID	Product Name	Corrected Product Name	Brand Name	Price ($)	Imputed Price	Quantity	Category	Imputed Categ
2	28-JAN-US	laptop	Laptop	Dell	$1,000.00	$1,000.00	30	Electronics	Electronics
3	15-FEB-US	Sneakers	Sneakers	Nike	$80.00	$80.00	15	Fashion	Fashion
4	03-MAR-US	Coffee Maker	Coffee Maker	Keurig	$130.00	$130.00	40	Kitchen	Kitchen
5	11-APR-US	smartphone	Smartphone	Samsung	$900.00	$900.00	25	Electronics	Electronics
6	22-MAY-US	Backpack	Backpack	North Face	$70.00	$70.00	20		Accessories

Summary: Formatting Price Column as Currency

I changed the data format of the Price column to Currency in Excel using the Number Format option. This displays all price values with the appropriate currency symbol and decimal places, improving readability and consistency.

● Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY "format



	D	E	F	G	H	I	J	K
1	Brand Name	Price ($)	Imputed Price	Quantity	Category	Imputed Category	Manufacturing Date	Country code
2	Dell	$1,000.00	$1,000.00	30	Electronics	Electronics	28-01-26	US
3	Nike	$80.00	$80.00	15	Fashion	Fashion	15-02-26	US
4	Keurig	$130.00	$130.00	40	Kitchen	Kitchen	03-03-26	US
5	Samsung	$900.00	$900.00	25	Electronics	Electronics	11-04-26	US
6	North Face	$70.00	$70.00	20		Accessories	22-05-26	US
7	Sony		$250.00	45	Electronics	Electronics	07-06-26	UK
8	Adidas	$30.00	$30.00	5	Fashion	Fashion	19-07-26	UK

Summary: Formatting Manufacturing Date Column

I used the LEFT and DATEVALUE functions to extract the date portion from the Product ID and convert it into a valid Excel date value using the year 2026. Then, I formatted the Manufacturing Date column as DD-MM-YYYY to maintain a consistent date format across the dataset.

6. Conditional Formatting:

● Apply data bar or color scales conditional formatting in the "Price" column.

Brand Name	Price ($)	Imputed Price	Quantity	Category
Dell	$1,000.00	$1,000.00	30	Electronics
Nike	$80.00	$80.00	15	Fashion
Keurig	$130.00	$130.00	40	Kitchen
Samsung	$900.00	$900.00	25	Electronics
North Face	$70.00	$70.00	20	
Sony		$250.00	45	Electronics
Adidas	$30.00	$30.00	5	Fashion
Ninja	$90.00	$90.00	35	Kitchen
Apple	$500.00	$500.00	50	Electronics
Timberland	$130.00	$130.00	10	Outdoor
HP	$950.00	$950.00	25	Electronics
Adidas	$90.00	$90.00	40	

Summary: Applying Conditional Formatting to Price Column

I applied Data Bars conditional formatting to the Price column in Excel to visually represent price variations. This helps easily identify higher and lower-priced products and makes data comparison easier.

● Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."

Brand Name	Price ($)	Imputed Price	Quantity	Category	Imputed Category	Manufacturing Date	Country code
Dell	$1,000.00	$1,000.00	30	Electronics	Electronics	28-01-26	US
Nike	$80.00	$80.00	15	Fashion	Fashion	15-02-26	US
Keurig	$130.00	$130.00	40	Kitchen	Kitchen	03-03-26	US
Samsung	$900.00	$900.00	25	Electronics	Electronics	11-04-26	US
North Face	$70.00	$70.00	20		Accessories	22-05-26	US
Sony		$250.00	45	Electronics	Electronics	07-06-26	UK
Adidas	$30.00	$30.00	5	Fashion	Fashion	19-07-26	UK
Ninja	$90.00	$90.00	35	Kitchen	Kitchen	23-08-26	UK
Apple	$500.00	$500.00	50	Electronics	Electronics	05-09-26	UK
Timberland	$130.00	$130.00	10	Outdoor	Outdoor	14-10-26	UK
HP	$950.00	$950.00	25	Electronics	Electronics	17-06-26	IN
Adidas	$90.00	$90.00	40		Footwear	25-11-26	AU
Nespresso	$120.00	$120.00	35		Kitchen	08-12-26	DE
Fitbit	$150.00	$150.00	15	Electronics	Electronics	18-02-26	CA

Summary: Conditional Formatting for Electronics Category

I created a custom conditional formatting rule in Excel to highlight cells in the Category column where the value is "Electronics". This makes it easier to identify electronics products and improves data visualization.